



Calculating the slit width and slit separation using
diffraction through a single slit and a double slit
at a 650nm wavelength light beam

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1 Introduction

The phenomena of light passing through slits of an opening, and examining how the light interacts with said slit has been documented for decades, with many physicists studying these phenomena to further develop accurate models of light. However, it was noticed that there was a difference between the patterns created from a single slit experiment and a double slit experiment.

The original double slit experiment was first performed by Thomas Young at the Royal Institute [4]. Young postulated that, if light was a wave, then it must obey the law of superposition and have constructive and destructive interference with other wave fronts (similar to sound waves and its superposition characteristics)[1]. These characteristics then proved true, as interference lines then appeared on the screen Young used in his double-slit experiment.

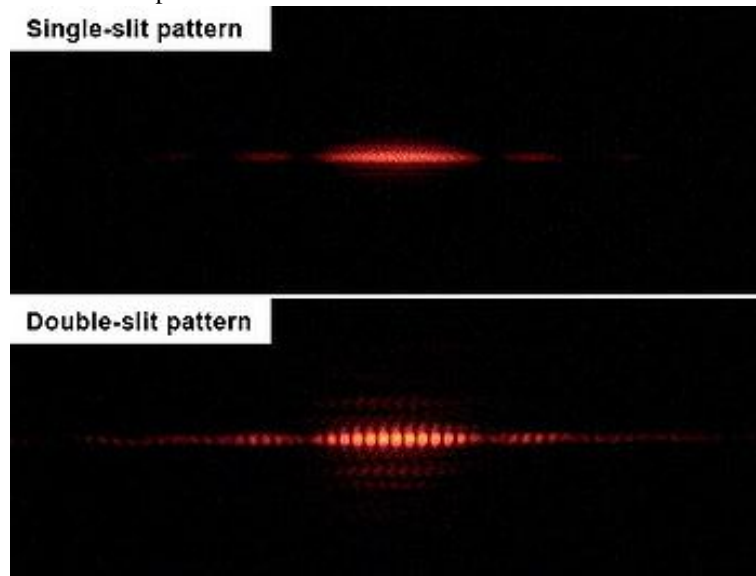


Image source: <https://www.universetoday.com/83380/double-slit-experiment/>

The image above shows the fringes of light on the bottom picture. These fringes occur due to the constructive and destructive interference that the light particles go through.

The application of this theory has been quite extensive. Ranging from the fields of astronomy, where spectrometers are used to discover the composition of stars[5], to laser technology on disks (like CDs and DVDs)[2]. It also has application in medical technology, such as enhancing x-rays using diffraction for images with less scattering [3].

The mathematical equations that will be used within this experiment are as follows:

$$a \sin(\theta) = m' \lambda \quad (1)$$

Where a is the slit width, θ is the angle between the central maxima and the n 'th

order maxima and λ is the wavelength. The next equation, used in the double slit, is:

$$d \sin(\theta) = m\lambda \quad (2)$$

where d is the distance between the slits in a double slit experiment, m is the order of the maxima being observed, and λ is the wavelength.

To calculate the angle used, we shall examine the distance from the central maxima to the first order maxima, and then also use the distance from the slit to the screen the diffraction patterns are shined upon. Then we apply these values to this formula:

$$\theta = \tan^{-1}\left(\frac{\text{distance from maxima}}{\text{distance to screen}}\right) \quad (3)$$

For this experiment, a light beam of 650 nm was used.

2 Aim

The aim of this experiment is to calculate the slit width of the single slit experiment, and to calculate the distance between slits in the double slit experiment. The values shall be calculated based on the theories of diffracting light as well as the associated mathematical models. Namely, equation (1), (2), and (3).

3 Method

For both of these procedures, an optic bench is utilised. An adjustable slit-disk is placed such that the light passes through the slits that are set up for the current experiment. The slit-disk mount must also be on rails such that it can be moved backwards and forwards.

On the opposite end, a light detector is mounted on another rail such that the 2 rails are perpendicular to one another (this rail shall move horizontally while the other moves vertically). The gain on the light detector can also be adjusted accordingly if need be.

3.1 Single slit experiment

1. Set the slit width to the 0.04mm width.
2. Move the light sensor to one side of the rail. This is to ensure that it fully captures the entire wave front spectrum that will be emitted without having to move the sensor back and forth.
3. Ensure that the computer is properly set up and connected with the sensor for optimal data collection.
4. Once all apparatus is ready, begin recording on the computer.
5. While the computer is recording data, move the sensor at a constant velocity across the rails. The sensor will detect the different intensities of the light.

6. Check on the computer so see if the data is recorded correctly. It should represent a series of peaks and plateau. The peaks signifying different orders of maximas and the plateau representing areas of low to no light intensity.
7. Repeat the process if needed.

3.2 Double slit experiment

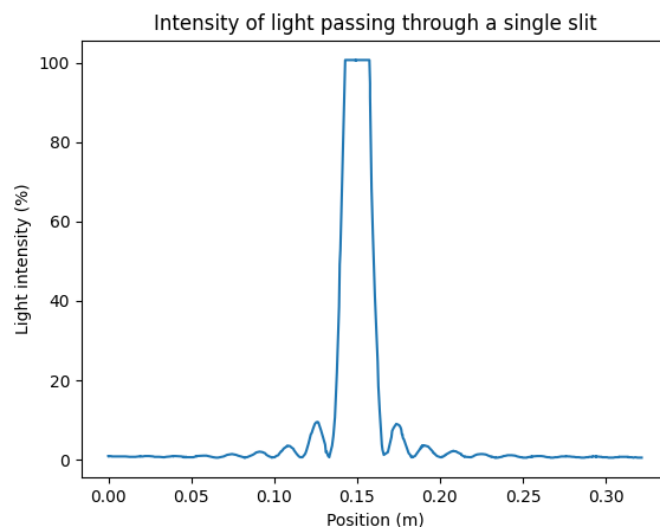
This procedure is very similar to the single slit experiment. The only difference being that the slit shall be moved to 2 distances. The first distance being 0.25m from the sensor, and the next being 0.5m from the sensor. The procedure will go as follows:

1. Change the slit disk into the multiple slit disk set. This multiple slit disk and a slit separation of 0.25mm and a slit width of 0.04mm.
2. Adjust the slit to the desired spot (either 0.25m or 0.50m).
3. Once again, move the light sensor from on end to the other.
4. Read the corresponding measurement on the computer.
5. Repeat this process at a distance of 0.50m

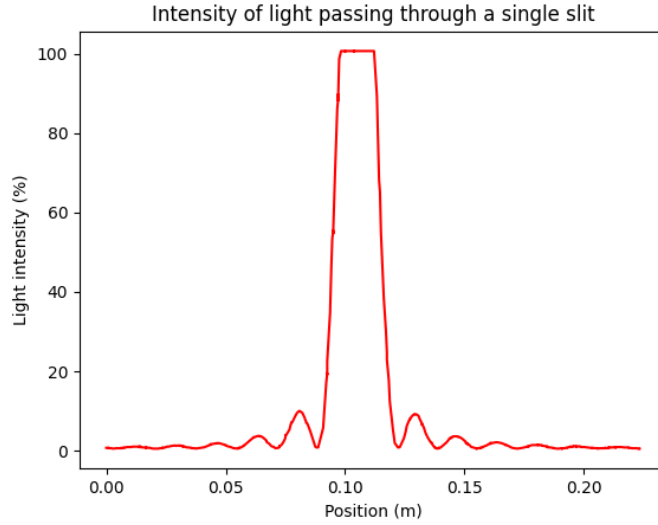
4 Results

4.1 Single slit experiment

This subsection focuses primarily on the single slit experiment and necessary calculations needed.



trial 1 of the single slit experiment



Trial 2 of the single slit experiment

For calculations, the first trial shall be used. The middle point of the central maxima is at approximately 0.15m. The distance to the first order maxima is approximately 0.174m and 0.126m. Thus, the distance between the maxima and the central maxima is approximately $0.174m - 0.150m = 0.024m$ and $0.150m - 0.126m = 0.024m$. The distance from the slit to the screen is 0.500m. Thus, the angle θ is:

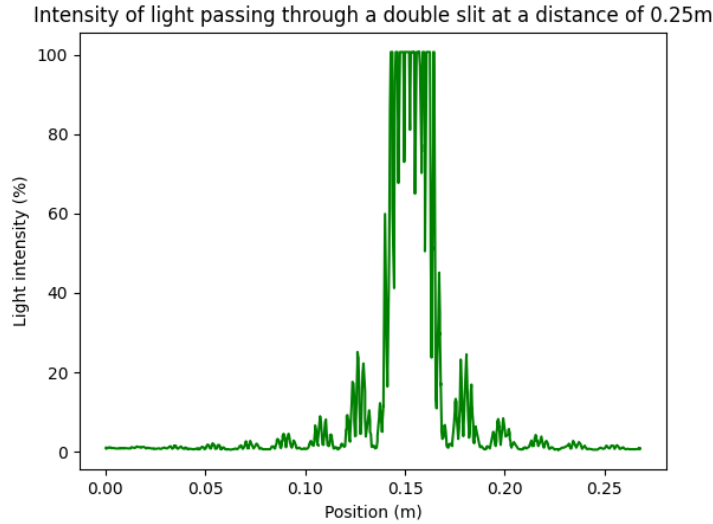
$$\theta = \tan^{-1}\left(\frac{0.024m}{0.500m}\right) = 2.748^{\circ}$$

Using this angle, we can calculate the slit width using the first equation:

$$a = \frac{650 \times 10^{-6}}{\sin(2.748)} = 0.0136mm$$

The actual slit width is 0.04mm. The error margin for the calculation is 66.107%.

4.2 Double slit experiment: distance of 0.250m



This graph, similar to the first, has a centre that is situated at approximately 0.150m. The centre of the left most first maxima is approximately 0.1264m, and the centre of the right most first maxima is approximately at 0.1808m. Thus, the distance between the central maxima and first order maxima is $0.1808m - 0.1500m = 0.0308m$ and $0.1500m - 0.1264m = 0.0236m$

For the first peak, the angle θ is:

$$\theta = \tan^{-1}\left(\frac{0.0236m}{0.25m}\right) = 5.393^\circ$$

And for the second peak, the angle θ is:

$$\theta = \tan^{-1}\left(\frac{0.0308m}{0.25m}\right) = 7.023^\circ$$

With these angles, we can now calculate the distance between the two slits. For the first angle, we have:

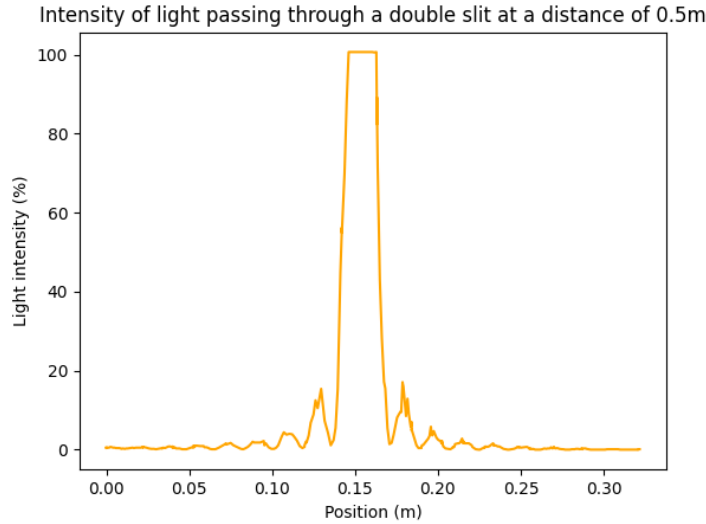
$$d = \frac{650 \times 10^{-6}}{\sin(5.393)} = 0.007mm$$

And when calculating with the 2nd angle we have that:

$$d = \frac{650 \times 10^{-6}}{\sin(7.023)} = 0.005mm$$

The actual slit separation is 0.25mm. The first has an error margin of 97.2% and the second is 98%

4.3 Double slit experiment: distance at 0.5m



This graph, similar to the first, has a centre that is situated at approximately 0.150m. The centre of the left most first maxima is approximately 0.1296m, and the centre of the right most first maxima is approximately at 0.1787m. Thus, the distance between the central maxima and first order maxima is $0.1787m - 0.1500m = 0.0286$ and $0.1500m - 0.1296m = 0.0204m$

For the first peak, the angle θ is:

$$\theta = \tan^{-1}\left(\frac{0.0286m}{0.25m}\right) = 6.526^\circ$$

And for the second peak, the angle θ is:

$$\theta = \tan^{-1}\left(\frac{0.0204m}{0.25m}\right) = 4.665^\circ$$

With these angles, we can now calculate the distance between the two slits. For the first angle we have:

$$d = \frac{650 \times 10^{-6}}{\sin(6.526)} = 0.006mm$$

And when calculating with the 2nd angle we have that:

$$d = \frac{650 \times 10^{-6}}{\sin(4.665)} = 0.008mm$$

The actual slit separation is 0.25mm. The first has an error margin of 97.6% and the second is 96.8%

5 Discussion

Regarding the single slit experiment, both graphs represent an expected light diffraction experiment. Both contain a central maximum with a large intensity of light, with smaller order maxima adjacent to the central maxima that decrease in intensity. Data collection appears as was theoretically described. The calculation of the slit width seems to be more inaccurate, however. The calculated slit width is 0.0136mm, which is not accurate or close to the actual width of 0.04mm. While it is only off by 66.107%, it is still a significant difference. This may be attributed to either incorrect data measurements, or incorrect interpolation of processed data.

Regarding each of the double slit experiments, graphically, the 0.25m distance graph seems far more erratic and seems to oscillate a lot more frequently than the other graphs. However, the overall theoretical "shape" of the experiment is visible. Reading data from this was difficult and may be the reason for the drastic differences in slit separation calculation and actual slit separation value, with all values being within the 90% range. While the graphed data does seem true to the theory, the calculated value seems to drastically disagree.

Overall, the data graphed does meet the theoretically described concepts, but the calculated values differ drastically from the actual values.

6 Conclusion

The aim of this experiment was to use the diffraction nature of light to calculate the slit width and slit separation of a single slit and double slit experiment. While the phenomena described within the background did occur, and the results graphed do match that of the theory, the calculated width and separation values were drastically off, with the closest calculation still being over 60% off. Because both the observed phenomena and graphed results match that of the theory, the calculated results must be an error due to either incorrect experimental procedure or incorrect data mining from the processed data or mathematical errors.

Note that this experiment does show great value in its application. Even though the aim of the experiment was to calculate slit width and slit separation, the entire process can have great use in fields such as optics and any field that uses lighting or any other electromagnetic spectrum manipulation (such as the medical field with applications in x-ray imaging). Knowing some factors, one can manipulate the others. Thus, this experiment can be optimised greatly in several directions depending on the individual and their current needs regarding their application.

Even though the calculated results were off by a significant factor, the theoretical model still appears to hold and the experiment (with further refinement in either procedure or carefulness regarding data and calculation) shows great potential.

7 References

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